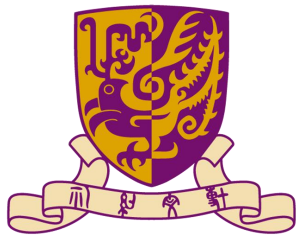
# DDA3020: Machine Learning - Tutorial 1

## - Python, Numpy, Scikit-Learn Basis

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Jan. 11&12, 2023



# Course Information

## Tutorial:

**Venue:** TC 103

**Time**: To be decided

### TAs:

**Junyi Ao (Lead):** junyiao1@link.cuhk.edu.cn, **OH:** Wed 4:30-5:30pm, RB311, H.L. Tu Building

**Haoying Li:** 222043004@link.cuhk.edu.cn, **OH:** Thu 2:30-3:30pm, Seat 38, SDS Lab, 4F, Zhixin Building

**Dan Qiao:** danqiao@link.cuhk.edu.cn, **OH:** Fri 4:00-5:00pm, Seat 81, SDS Lab, 4F, Zhixin Building

**Sho Inoue:** shoinoue@link.cuhk.edu.cn, **OH:** Thu 3:30-4:30pm, Seat 82, SDS Lab, 4F, Zhixin Building

**Xiaoyu Peng:** 118010234@link.cuhk.edu.cn, **OH:** Fri 2:00-3:00pm, location (TBD)

### UStaff-TAs (Appoint via email or Wechat group in advance):

**Yifan Wang:** 119010317@link.cuhk.edu.cn, **OH:** Mon 14:00 - 15:00pm, start-up zone library

**Rongxiao Qu:** 120020144@link.cuhk.edu.cn, **OH:** Tue 9:00 - 10:00am, start-up zone library

**Dongze Wu:** 119010337@link.cuhk.edu.cn, **OH:** Fri 10:30 - 11:30am, start-up zone library

**Nan Xiao:** 119010344@link.cuhk.edu.cn, **OH:** Thu 20:00 - 21:00pm, start-up zone library

# Python Packages for ML

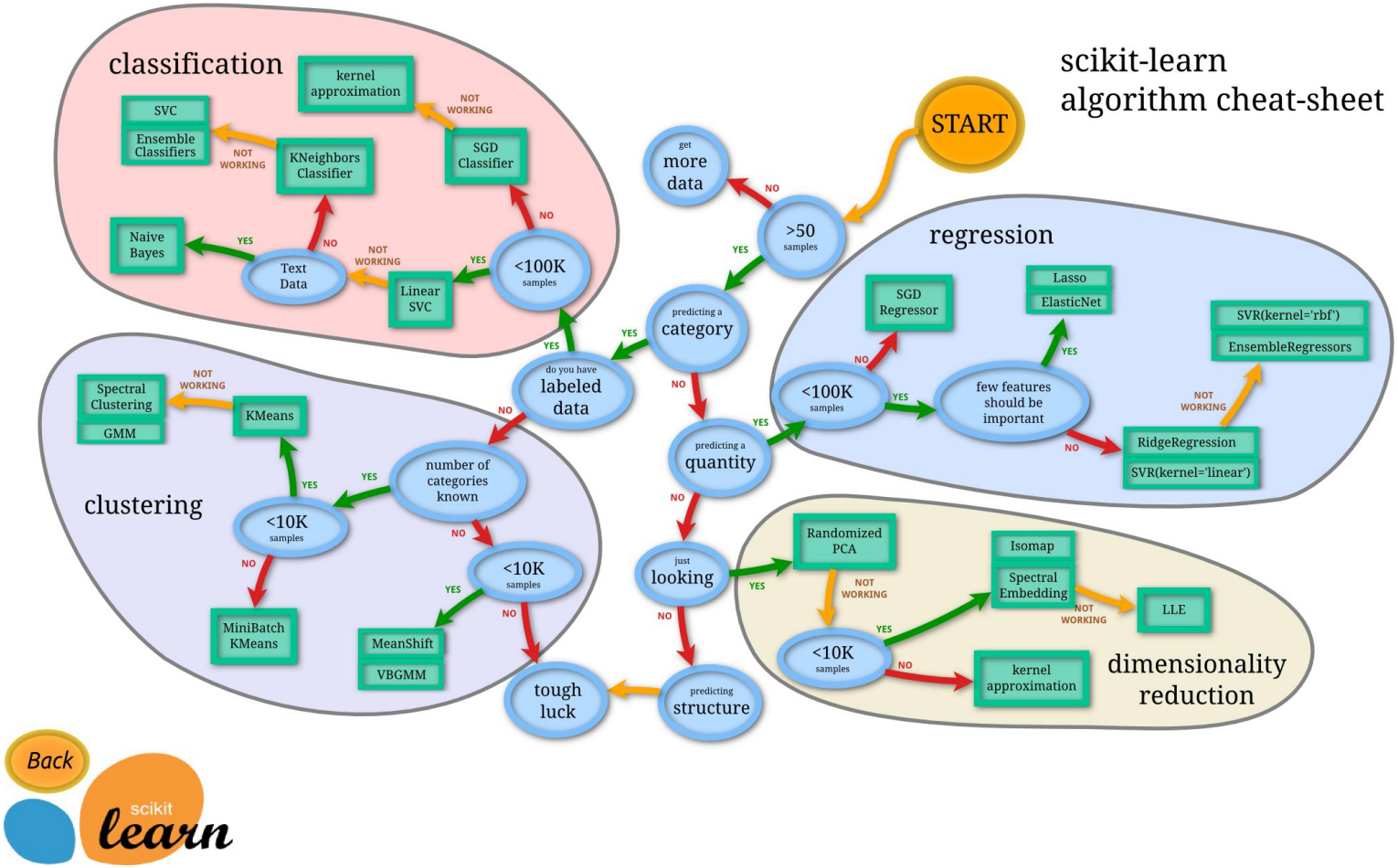
- Numpy + Scipy:
  i)Useful data structure: ndarray, matrix, array
  ii)Algorithms: statistics, scientific computation
- Pandas:
  Processing table-like dataset (such as Excel), DataFrame, Series
- Scikit-learn: Various data mining/machine learning functions
- NLTK: Natural Language Toolkit
- Visualization: Matplotlib, Seaborn...
- Deep Learning: TensorFlow (Google), PyTorch (Facebook), Caffe, Keras, MXNeT (Amazon)

Recap the lecture content, we can express the data in vector/matrix formula: one data sample with d features $\boldsymbol{x}_i \in R^d$, total n sample we can stack a matrix $X = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n] \in R^{d \times n}$

In this tutorial, we will briefly introduce the **Numpy: ndarray** and **Scikit-learn**.

# Scikit-Learn: Various data mining/machine learning functions

Strongly recommend to ref the Scikit-Learn official website:
https://scikit-Learn.org/stable/



## Tips On Studying This Course

1. Remember all the basic concepts in your course materials (e.g. Entropy; KL-divergence; Bias-Variance Trade-off)

2. Comprehend all the mathematical derivations and be able to derive them by yourselves. (e.g., SVM, K-means, GMM, EM)

3. Practice your coding skills often!
   Numpy, Pandas, Scikit-learn packages in python are very important!

4. Submit a comprehensive written report and a well-structured coding report (coding report is important!) in each homework

5. DO NOT CHEAT! (Coding similarity check will be conducted at the end of the semester)

# Demonstration on installing Anaconda

Move to Jupyter Notebook